Blockchain-Enabled Cloud Infrastructure Management for Trustworthy Autonomous Operations

Programme Code - Research in Computing CA1
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1 Research Problem Background

The rapid growth of cloud computing has enabled it to provide elastic, on-demand computing capacity, supporting businesses from banking to medicine (Zhang et al., 2022). Classical cloud infrastructure management systems are, however, faced with dominant challenges, primarily ensuring trust, security, and autonomous execution. These are attributed to centralized control, low transparency, and susceptibility to cyberattacks, which compromise system reliability and users' confidence (Fernández-Caramés & Fraga-Lamas, 2020).

Blockchain technology has been considered a potential solution to such issues by providing a decentralized, tamper-evident, and unchangeable ledger. The integration of blockchain with cloud infrastructure management makes it more transparent, strong, and tamper-proof (Singh et al., 2023). Blockchain eliminates the point of failure of traditional cloud management through the dispersal of control over multiple nodes, thereby making it more secure and reliable (Tian et al., 2021).

Autonomous execution is one of the key attributes of cloud infrastructure management, wherein artificial intelligence (AI) and machine learning (ML) technologies are employed for automating service orchestration, fault detection, and resource assignment (Gupta et al., 2022). Recent AI-driven cloud management tools, however, remain unverifiable and reliable in that decisions tend to be typically taken by black-box algorithms that run on centralized systems (Rahman et al., 2021). The application of blockchain technology has the potential to make the system more reliable through offering transparency in every decision that is made in operations and confirmation in a decentralized process (Li et al., 2023).

Moreover, smart contracts—self-executing contracts deployed on a blockchain—can also make most things about cloud management automated, such as provisioning, security enforcement, and monitoring compliance (Xu et al., 2022). The contracts eliminate human error, minimize errors, and enforce pre-specified policies in the same way each time. With these benefits aside, deployment of blockchain for cloud infrastructure management is difficult because of scalability, interoperability, and computational overhead (Apostolopoulos et al., 2023).

There have been some recent investigations of hybrid blockchain architectures and off-chain scaling strategies to mitigate these limitations without sacrificing security and decentralization (Hameed et al., 2023). Besides, studies are being conducted on the design of energy-efficient consensus protocols specific to cloud environments (Sharma et al., 2022).

Finally, blockchain and cloud computing convergence has the potential to enhance trust and security in self-managed cloud operation. However, more research must be carried out to scale the use of blockchain, assess performance trade-offs, as well as design frameworks for incorporation in existing cloud management systems. This research strives to fill these gaps and contribute towards the development of a trustworthy blockchain-based cloud infrastructure.

2 Research Question

How can blockchain technology be utilized to the best of its ability in cloud infrastructure management to gain increased trust, security, and self-operation at the cost of scalability and performance efficiency?

The research question places stress on the quantitative dimensions of blockchain-based cloud infrastructure, i.e.,

Trust and Security – Quantifying the extent to which blockchain ensures higher transparency, access control, and data integrity within the cloud.

Autonomous Operations – Examining the potential of decentralized decision-making and smart contracts in automating cloud management.

Scalability and Performance – Evaluating the influence of blockchain support on system latency, resource usage, and processing overhead in cloud operations.

By addressing these issues, the research will provide a systematic framework for using blockchain in cloud infrastructure management and make it practicable.

3 Justification

Blockchain application to cloud infrastructure management is significant due to increasing need for secure, transparent, and autonomous cloud operation. Cloud management platforms are susceptible to security attacks, misuse, and operational inefficiency (Singh et al., 2023). Blockchain, through its decentralized process, establishes trust through offering immutable records and tamper-proof access control, addressing primary challenges in cloud computing (Tian et al., 2021).

This research becomes possible due to the availability of blockchain platforms like Hyperledger Fabric and Ethereum, and cloud service providers providing Blockchain-as-a-Service (BaaS) like AWS and Azure (Li et al., 2023). Performance metrics like transaction latency, throughput, and security resilience provide measurable insights into how blockchain affects cloud management (Gupta et al., 2022).

The research is critical since it presents a structured model of blockchain-supported cloud management that affects industry implementation and offers efficiency and scalability. The ethical implications are data confidentiality, decentralization, and greenness, particularly with energy-conscious consensus algorithms (Apostolopoulos et al., 2023). From consideration of the above implications, this research provides the ability to build a safe, secure, and self-dependent cloud infrastructure to be beneficial for companies, service providers, and end-users.

4 Specific Items to Be Addressed

The major highlight of this research shall be the study of various major issues throughout the field of blockchain-based cloud management:

Security and Trust – Study of how data integrity, access control, and tamper resistance are enabled by blockchain in clouds (Singh et al., 2023).

Autonomous Cloud Operations – Analysis of how distributed rulemaking and intelligent contracts can implement automated resource provisioning, rule-based compliance, and fault tolerance (Xu et al., 2022).

Performance and Scalability – Defining the impact of blockchain on latency, throughput, and system overhead and what are the likely mechanisms for optimization (Hameed et al., 2023).

Challenges in Integration – Implementing interoperability, computation efficiency, and implementation issues within hybrid cloud-blockchain platforms (Sharma et al., 2022).

Ethical and Sustainability Considerations – Determining privacy issues, decentralized decision-making, and energy-efficient consensus methods (Apostolopoulos et al., 2023).

By resolving these aspects, the study intends to formulate a model for an efficient and scalable blockchain-supported cloud infrastructure.

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